AUSTIN AIRBNB MARKET

Stop Chasing High Nightly Rates. Start Chasing Occupancy.

TO: Austin Airbnb Investor
FROM: Wake Forest Analytics Team
DATE: March 5, 2026
SUBJECT: How Mid-Range Entire Homes Deliver the Strongest Risk-Adjusted Return in Austin

Executive Summary

The Austin short-term rental market is large, liquid, and deeply segmented. With nearly 6,000 active listings spanning radically different price points and occupancy profiles, the question for a prospective investor is not whether Austin is a viable Airbnb market. It clearly is. The question is which segment delivers the most predictable, risk-adjusted return. Our analysis of 5,835 listings using K-Means clustering and XGBoost regression provides a clear answer: mid-range entire homes (Cluster 0) averaging $174 per night achieve a 34% booking rate, the highest of any segment, generating approximately $21,600 in annual gross revenue with far lower vacancy exposure than the premium tier.

The path to maximising that return is equally clear. Bedrooms, bathrooms, and neighbourhood selection are the dominant price drivers. A single targeted upgrade, adding a hot tub, lifts average nightly rate by nearly $200 and pushes annual revenue toward $26,000 with minimal incremental risk. We recommend acquiring a two-bedroom entire home in a central Austin neighbourhood, pricing at $150 to $200 per night, and pursuing Superhost status from day one. This strategy captures the broadest demand pool in the market while preserving meaningful upside through dynamic pricing during Austin's high-demand event calendar.

Part II: Project Deployment Outline

Value

Every investment decision starts with a question worth answering precisely. Here, the investor needs to know not just that Austin is a strong short-term rental market, but which property type, in which segment, at which price point, offers the most defensible return. Generic market averages cannot answer that. A framework that segments the market and quantifies the drivers of nightly price can. The deployment goal is to identify the highest-return property segment and deliver a listing strategy that maximises both nightly rate and occupancy in a single, coherent recommendation.

Target

To deliver that recommendation, the analysis pursues two linked objectives. First, K-Means clustering groups the 3,130 model-ready listings into distinct segments based on price, size, room type, and booking activity, producing an investor-ready market map without requiring a labelled outcome. Second, XGBoost regression predicts log(nightly price), the continuous target that determines revenue potential. The log transformation was applied because raw price exhibited severe right skew (skewness = 8.5), which would inflate error metrics and distort model performance. A secondary binary target, whether a listing is effectively booked defined as 90-day availability below 40 days, informs the occupancy dimension of the investment recommendation.

Performance

With objectives defined, the next question is how to know when the models are working. Clustering was evaluated using the elbow plot and silhouette score across k = 2 through 6. The silhouette score measures how well each listing fits its assigned cluster relative to neighbouring clusters, with higher values indicating cleaner separation. For regression, the primary metric is Test Adjusted R^2, selected because it penalises

model complexity and reflects generalisation to unseen listings. MAPE provides a scale-independent accuracy measure, directly interpretable in dollar terms against segment nightly averages. Throughout, we prioritise test performance over training performance to guard against overfitting.

Fuel

Before any model could run, the data needed to be made trustworthy. The raw dataset contained 5,835 listings and 54 variables. Three preparation decisions shaped the final modelling dataset. First, columns with prohibitive null rates were dropped: square footage (94.8% missing), weekly price (61.8%), and security deposit (52.5%). Second, price and fee fields stored as strings were converted to numeric values, with null cleaning fees imputed as zero to reflect no stated charge. Third, rows missing values on core modelling variables were removed and nightly prices above the 99th percentile ($1,300) were capped, yielding 3,130 clean observations.

Exploratory analysis then surfaced features worth engineering explicitly. Hot tub listings averaged $472 per night versus $274 for listings without one, a $198 gap significant enough to warrant a dedicated binary flag. Binary flags for pool, washer, gym, and pets allowed were created on the same basis. A booked indicator was constructed by classifying listings with fewer than 40 days available over the next 90 as effectively occupied. These decisions were not arbitrary. Each was grounded in observed price differentials from the data itself.

Algorithm

With clean data and well-defined targets, the analysis moves to the core modelling work. Two methods were applied in sequence, each building on the other to produce a unified investment recommendation.

Method 1: K-Means Clustering

The first task was to map the market. Clustering variables included price, bedrooms, bathrooms, accommodates, room type (label encoded), review score rating, and 90-day availability. All variables were standardised before clustering to prevent scale dominance. We tested k = 2 through 6, evaluating each using both the elbow plot and silhouette score.

The silhouette score peaked at k = 2 (0.357) but two clusters produced an overly broad split, separating only cheap listings from expensive ones with no actionable nuance for an investor. At k = 3 (silhouette = 0.299), three structurally meaningful segments emerge with distinct pricing, size, and occupancy profiles. Scores flatten beyond k = 3, confirming that additional clusters fragment the market without adding interpretive value. We selected k = 3 as the optimal solution on both statistical and business grounds.

Cluster	Avg Price	Avg Beds	Accommodates	Booking Rate	Count
0	$174	1.4	4 guests	34%	1,581
1	$479	3.3	8 guests	22%	726
2	$85	1.0	2 guests	23%	823

⋆ **Cluster 0** = Mid-Range Homes **Cluster 1** = Premium Large Homes **Cluster 2** = Budget Private Rooms

Cluster 0 is the recommended investment target. It delivers the highest booking rate in the dataset (34%), the largest addressable market (1,581 listings), and consistent nightly revenue at $174. Cluster 1 commands higher per-night revenue ($479) but sits vacant 78% of the time, making it viable only for investors willing to absorb event-driven demand variability. Cluster 2 offers neither competitive pricing nor meaningful occupancy and is not recommended.

With a clear picture of where the market segments lie, the natural next question is what drives price within and across those segments. That is where regression takes over.

Method 2: XGBoost Regression

Knowing that Cluster 0 is the right segment tells the investor where to play. Knowing what drives price within that segment tells the investor how to win. Five models were trained on an 80/20 train-test split with log(price) as the target. Input variables included bedrooms, bathrooms, accommodates, room type, property type, neighbourhood, review score, cleaning fee, amenity flags, host superhost status, instant bookable, minimum nights, and cluster label. The final XGBoost model was tuned using Optuna Bayesian optimisation across 100 trials, yielding the parameter set that maximised held-out R^2.

Model	Train R^2	Test Adj R^2	MAPE
Log Linear Model	0.714	0.622	33.1%
Random Forest	0.958	0.613	36.4%
Base XGBoost	0.910	0.652	32.4%
Tuned XGBoost (Optuna)	**0.787**	**0.671**	**31.5%**
Tuned XGBoost with New Features	0.807	0.660	32.1%

The tuned XGBoost model is selected as the final model. It achieves the strongest Test Adjusted R^2 (0.671) and lowest MAPE (31.5%), with an overfitting gap of 0.116 that is well within acceptable bounds compared to Random Forest (0.345). The remaining unexplained variance of approximately 33% reflects genuinely unobservable factors including listing photo quality, seasonal demand concentration, and individual host pricing discretion. These are structural limitations of the dataset, not model failures.

Feature importance confirms that bedrooms rank first (0.221), cluster label second (0.150), and bathrooms third (0.073). The prominence of cluster label in the feature importance output directly validates the segmentation work. A property's market positioning is nearly as predictive of its nightly rate as its physical size, which means the segmentation and the pricing model are telling a consistent story.

Launch

The models have spoken. What follows translates their output into a concrete action plan the investor can execute.

Primary Recommendation

Acquire a two-bedroom entire home accommodating three to four guests in a central Austin neighbourhood such as Zilker, Travis Heights, or East Riverside. Price at $150 to $200 per night. This is Cluster 0 territory and it represents the strongest entry point in the Austin market on a risk-adjusted basis, combining the highest booking rate (34%), the largest and most liquid market (1,581 listings), and consistent monthly cash flow.

Pricing Strategy

- Anchor the base nightly rate at $174, the Cluster 0 observed average, and adjust dynamically using tools such as PriceLabs or Beyond to capture premiums during ACL Festival, SXSW, and Formula 1 weekend.
- Prioritise a hot tub installation as the first capital upgrade. Listings with hot tubs average $472 per night versus $274 without, a $198 uplift translating to approximately $4,400 in additional annual revenue at observed occupancy rates.
- Pool installations offer a more modest premium of approximately $33 per night and should be considered a secondary upgrade only after the hot tub.

Operations and Host Setup

- Enable instant booking and cap minimum stay at three nights to maximise the available booking window and reduce calendar gaps.

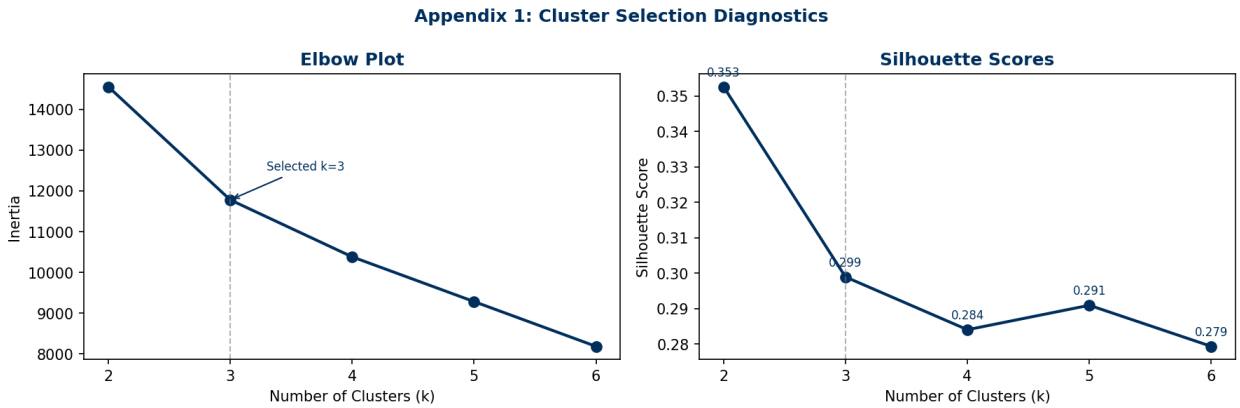
- Invest in guest experience from the first booking. Review scores across cleanliness, communication, and accuracy directly support search placement and repeat demand.
- Achieve Superhost status within the first six months. It improves search visibility and reduces booking friction for first-time guests.

Expected Financial Impact

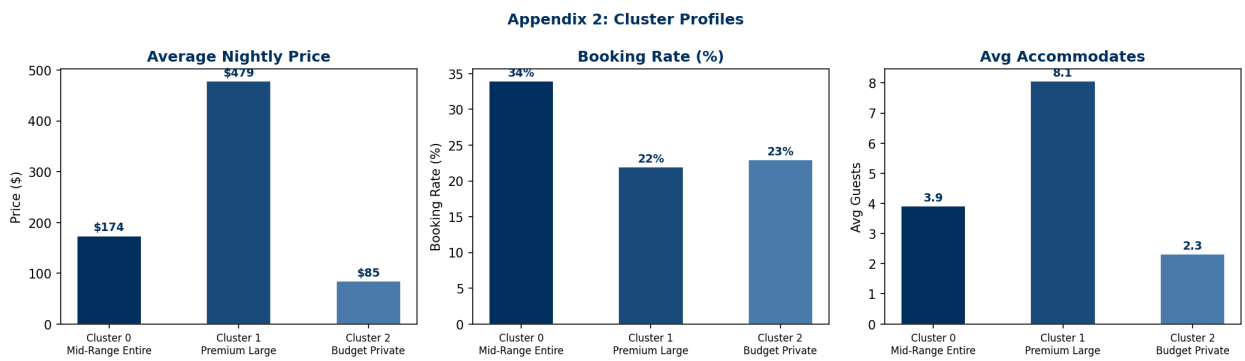
At Cluster 0 observed performance, a typical property generates approximately $21,600 in annual gross revenue ($174 per night at 34% occupancy across 365 days). A hot tub upgrade pushes this figure to approximately $26,000. A premium Cluster 1 property at equivalent occupancy would yield $38,500 annually but the 78% vacancy rate makes that scenario highly unlikely in practice. For an investor prioritising consistent monthly income over speculative upside, Cluster 0 is the structurally superior choice.

Part III: Appendix

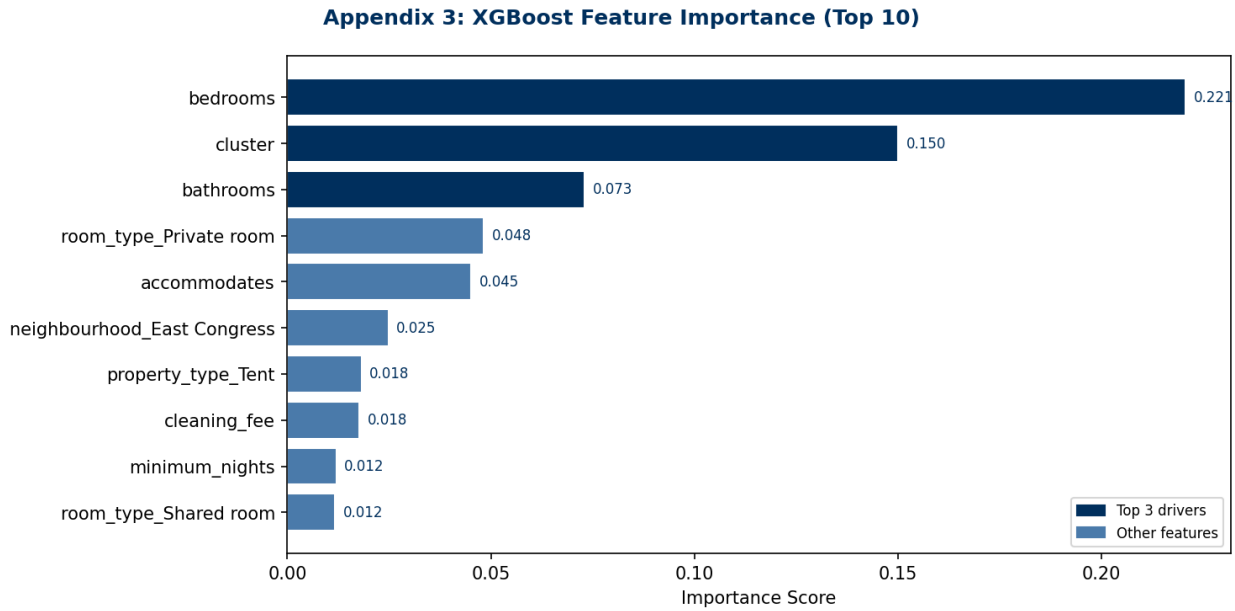
Appendix 1. Cluster Selection Diagnostics



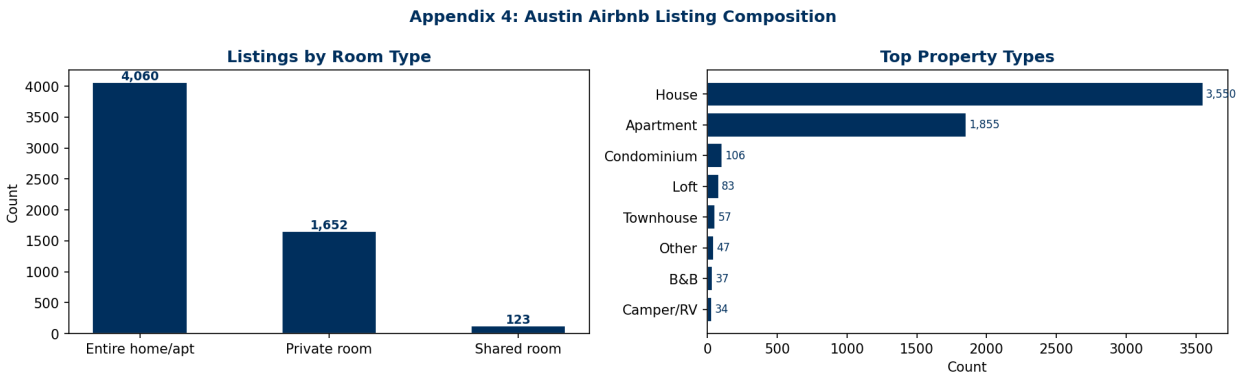
Appendix 2. Cluster Profiles



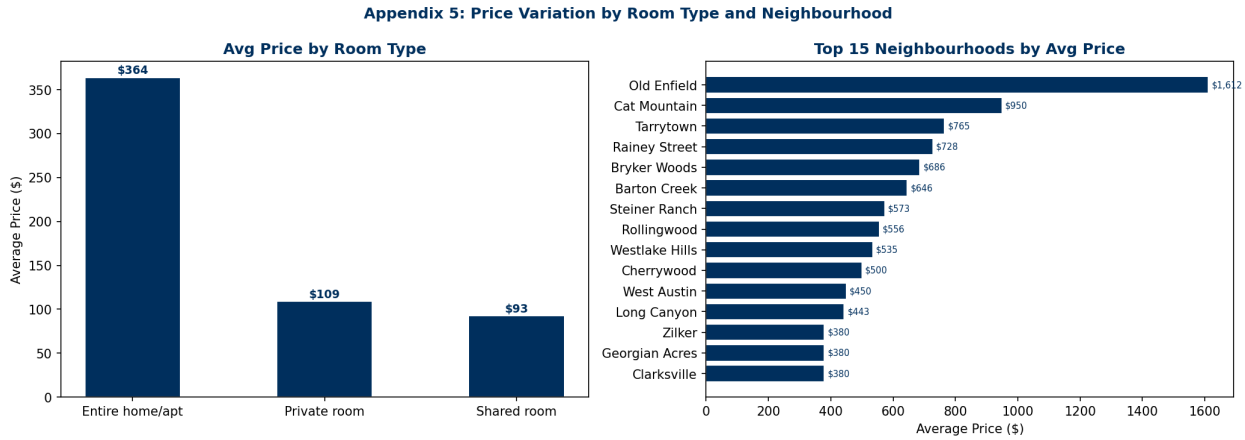
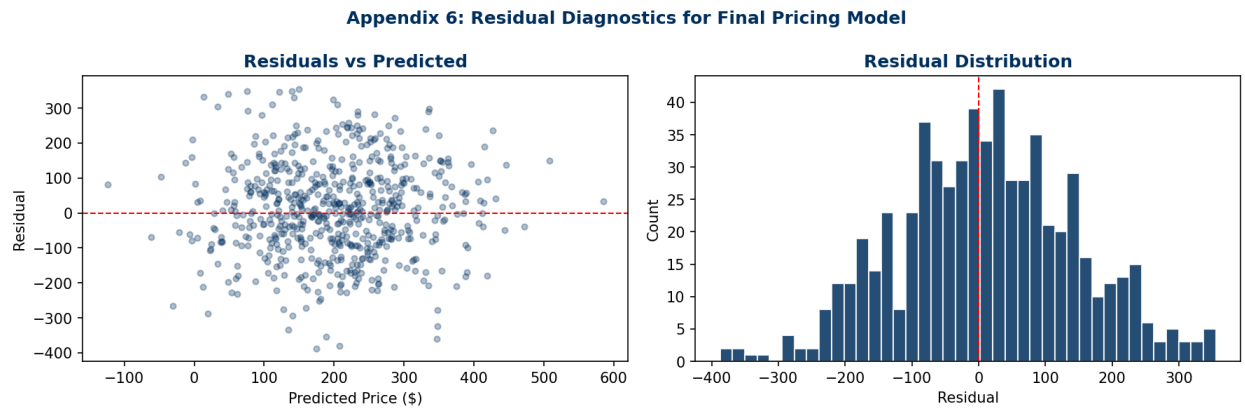
Appendix 3. XGBoost Feature Importance



Appendix 4. Austin Airbnb Listing Composition



Appendix 5. Price Variation by Room Type and Neighbourhood



Appendix 6. Residual Diagnostics for Final Pricing Model